1 Forward, Futures & Hedging

Formulas

$$\sum_{i=0}^{n-1} ar^i = \frac{a(1-r^n)}{1-r}$$

For given rate $r\%$ over time period $t \in [0, T]$, with annual period P, discount factor $d_{0,T}$ is given

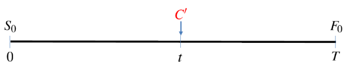
$$d_{0,T} = \left(1 + r\% \times \frac{T}{P}\right)^{-1} < 1$$

No Arbitrage Argument

To prove $A = B$, show strategies for $A \neq B$. If $A > B$, short on A, long on B, and vice versa.

1.1.2 Asset w/ Predictable Income

$$F_0 = (S_0 - C)/d_{0,T}, C = C' d_{0,t}$$

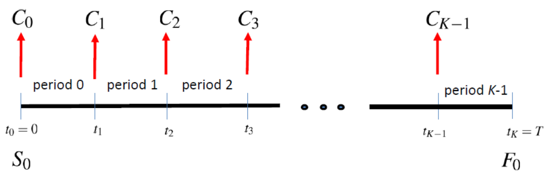


1.1.3 Asset w/ Continuous Dividend Yield

Annualized continuous dividend yield of q

$$F_0 = S_0 \exp((r-q)T)$$

1.1.4 Asset w/ Storage Costs

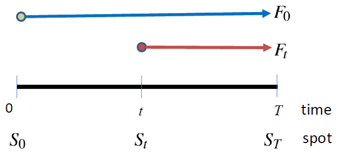


$$d_{0,K} F_0 = S_0 + \sum_{j=0}^{K-1} d_{0,j} C_j$$

1.1.5 Convenience Yield

$$d_{0,K} F_0 = S_0 + \sum_{j=0}^{K-1} d_{0,j} (C_j - y)$$

1.2 Value of Forward



Value of **long** position:

$$f_t = (F_t - F_0) d_{t,T}$$

1.3.1 Marked-to-Market Mechanism

1. Investor deposits intial margin
2. Account is marked to market each day on gain/loss of futures position
3. Margin account balance flutuates day to day
4. If balance drops below threshold, investor required to top up with cash

1.3.2 Futures-Forwards Equivalence

If interest rates are known to follow expectation dynamics, then theoretical futures and forward prices of corresponding contracts are identical.

1.4.1 Short Hedge

To **lock in** a selling price, applicable when

- already owns asset and expects to sell in futures
- does not own asset but will in future, and sell later afterward

Net cash inflow of $F_0 = S_T + (F_0 - S_T)$ is locked in.
With basis risk,

$$F_1 + b_2 = S_2 + (F_1 - F_2)$$

1.4.2 Long Hedge

To **lock in** a purchase price, applicable when hedger has to purchase asset in the future.
Net cash outflow of $F_0 = S_T - (S_T - F_0)$ is locked in.
With basis risk,

$$F_1 + b_2 = S_2 - (F_2 - F_1)$$

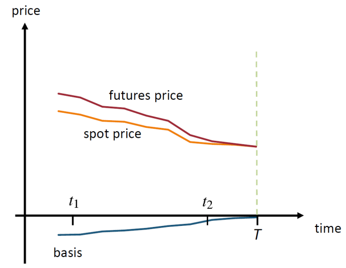
1.4.3 Basis Risk

Perfect hedges often not possible in reality.

1. unavailability of futures contract matching asset obligation dates
2. quantity of asset obligated may not be integer multiple of contract size
3. lack of liquidity in futures market
4. terms of delivery may not coincide with obligation
5. unavailability of futures contracts with underlying asset exactly matching asset hedged

Define basis b_i at time t_i, with spot price S_i and futures price F_i as

$$b_i = S_i - F_i$$



Effective price can be written as

$$F_1 + b_2 = S_1 - (b_1 - b_2)$$

Price at time t_1, offset by diff btw **initial basis** b_1 and **cover basis** b_2.
Should choose futures contract with maturity close, but later than expiration of hedge.

1.4.4 Cross Hedging

Futures and spot price of proxy asset at time t_i denoted by F_i^* and S_i^* respectively. Effective price achieved is

$$S_2 + (F_1^* - F_2^*) = F_1^* + (S_2^* - F_2^*) + (S_2 - S_2^*)$$

1.5 Minimum Variance Hedge

At t_1, hedger has obligation to purchase Q_a units of asset at time t_2. Cash flow at t_2 is

$$x = Q_a(S_1 + \Delta S)$$

Suppose F is futures price of contract used as hedge, Q_f denotes size of long futures contract taken. Hedge ratio

$$h = \frac{Q_f}{Q_a}$$

At time t_2, cash flow on hedged position

$$y = Q_a(S_1 + \Delta S - h\delta F)$$

$$\Delta F = F_2 - F_1$$

Minimum-Variance Hedge ratio

$$h^* = \frac{cov(\Delta S, \Delta F)}{var(\Delta F)}$$

$$h^* = \rho \frac{\sigma_{\Delta S}}{\sigma_{\Delta F}}$$

$$Q_f^* = Q_a h^*$$

2 Forwards & Futures in Practice

2.1.1 US Treasury Bills

- **Dollar price**: Typically quoted per \$100 of face value
- Often given as **discount yield** d

$$d = \frac{Par - P}{Par} \times \frac{360}{D}$$

where D is days to maturity.

- Price is then

$$P = Par \times \left(1 - d \times \frac{D}{360}\right)$$

- True yield is the **coupon equivalent yield** (CEY) or **investment rate**

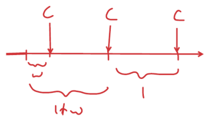
1. T-bills not more than half-year to maturity

$$i = \frac{100 - P}{P} \times \frac{y}{t}$$

2. T-bills more than half-year to maturity. Root of

$$\left(\frac{t}{2y} - 0.25\right) i^2 + \frac{t}{y} i + \frac{P - 100}{P} = 0$$

2.1.2 Accrued Interest



Bond price a.k.a. **full, dirty, or invoice price**

$$P = \frac{F}{\left(1 + \frac{r}{m}\right)^{w+n-1}} + \sum_{i=0}^{n-1} \frac{C}{\left(1 + \frac{r}{m}\right)^{w+i}}$$

Quoted prices a.k.a. **clean or flat price** does not include accrued interest.

$$DPrice = CPrice + AI$$

2.2.1 Repurchase Agreements

- Repurchase agreements = **repo** = **collateralized loan.**
- Underlying security held as **collateral** by counter-party.
- Counter-party entered **reverse repo**.
- Rate involved is **repo rate**.
- **Simple interest** rates. actual/360 in US.

1. At time $t = 0$, trader finances purchase of security worth S_0.
2. Enter T-day repo with Mr. Y.
3. Mr. Y demands haircut of $h\%$ and repo rate r.
4. Mr. Y lends $S_0(1 - h\%)$.
5. Trader has net outflow of $S_0 \times h\%$.
6. At time $t = T$, trader pays $F = S_0(1 - h\%)(1 + r \times T/360)$.
7. Trader receives security.
8. Equivalent to forward purchase of security at forward price F.
9. If security price increases to S_T then trader earns profit.
10. To carry out bet on decline using reverse repo, buy at S_0, sell at $S_0(1 - h\%)$, buy back at S_T.

2.2.2 SOFR

Secured Overnight Financing Rate (SOFR). Day count: actual/360. Compound SOFR Formula

$$r = \left[\prod_{i=1}^{T}\left(1 + \frac{r_i n_i}{Y}\right) - 1\right] \frac{Y}{d_c}$$

Simple SOFR Formula

$$r = \frac{1}{d_c} \sum_{i=1}^{T} r_i n_i$$

2.2.3 1-Month and 3-Month SOFR Futures

- 1-month SOFR Futures: \$5m loan for 1-month
- 3-month SOFR Futures: \$1m loan for 3-months
- SOFR Futures follow 30 day periods
- Quoted Futures price 1-month: $100 \times (1 - E(r_m))$, where r_m is the simple average SOFR for contract month
- Final Settlement price 1-month: $100 \times (1 - r_m)$.
- Quoted Futures price 3-months: $100 \times (1-$ expected compound average SOFR for contract period)
- Long position gains when expected SOFR falls (futures rises).
- Take up long position to hedge against rate falls.
- Long position is equivalent to agreeing to lend at rate

$$(100 - F_0)\%$$

- Short position gains when expected SOFR rises (futures falls).
- Take up Short position to hedge against rate rises.
- Short position is equivalent to agreeing to borrow at rate

$$(100 - F_0)\%$$

Changes to margin account

- 1-month: Decrease in futures price by 0.01 (rate rise by 1 bp) results in long position being decreased by \$41.67.
- 3-months: Decrease in futures price by 0.01 (rate rise by 1 bp) results in long position being decreased by \$25.